

EXPERIMENT - 2

Aim :

Convert the ER Diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

SQL QUERY

CREATE DATABASE IndianEcommerce ;

USE IndianEcommerce;

--1. Customer

CREATE TABLE Customer (

CustomerID INT PRIMARY KEY,

FirstName VARCHAR (50) NOT NULL,

LastName VARCHAR (50) NOT NULL,

Email VARCHAR (100) NOT NULL UNIQUE,

RegistrationDate DATE NOT NULL

);

--2. CUSTOMER_PHONE

-- Multivalued attribute of Customer

~~CREATE TABLE CustomerPhone (~~

~~CustomerID INT NOT NULL,~~

~~PhoneNumber VARCHAR (15) NOT NULL,~~

~~PRIMARY KEY (CustomerID)~~

~~REFERENCES Customer (CustomerID)~~

~~ON DELETE CASCADE~~

~~);~~

-- 3. SELLER

```
CREATE TABLE Seller (  
    SellerID INT PRIMARY KEY,  
    SellerName VARCHAR(100) NOT NULL,  
    Email VARCHAR(100) NOT NULL UNIQUE,  
    Phone VARCHAR(15) NOT NULL,  
    GSTIN VARCHAR(20) UNIQUE,  
    Rating DECIMAL(2,1),  
    CHECK (Rating >= 0 AND Rating <= 5)  
);
```

-- 4. CATEGORY

```
CREATE TABLE Category (  
    CategoryID INT PRIMARY KEY,  
    CategoryName VARCHAR(100) NOT NULL UNIQUE,  
    Description VARCHAR(255) .  
);
```

-- 5. PRODUCT

CREATE TABLE Product (

ProductID INT PRIMARY KEY,

ProductName VARCHAR(150) NOT NULL,

Price DECIMAL(10,2) NOT NULL,

Description VARCHAR(500),

Stock INT NOT NULL DEFAULT 0,

SellerID INT NOT NULL,

CategoryID INT NOT NULL,

FOREIGN KEY (SellerID)

REFERENCES Seller (SellerID)

ON DELETE CASCADE,

FOREIGN KEY (CategoryID)

REFERENCES Category (CategoryID)

ON DELETE CASCADE,

~~CHECK (Price >= 0),~~

~~CHECK (Stock >= 0)~~

);

--6. PRODUCT ELECTRONIC

-- Product Specialization

```
CREATE TABLE ElectronicProduct (  
    ProductID INT PRIMARY KEY,  
    Brand VARCHAR (100) NOT NULL,  
    Model VARCHAR (100) NOT NULL,  
    WarrantyMonths INT,  
  
    FOREIGN KEY (ProductID)  
        REFERENCES Product (ProductID)  
        ON DELETE CASCADE,  
  
    CHECK (WarrantyMonths >= 0)  
);
```

--7. CLOTHING PRODUCT

-- Product Specialization

```
CREATE TABLE ClothingProduct (  
    ProductID INT PRIMARY KEY,  
    Size VARCHAR (20) NOT NULL,  
    Material VARCHAR (100),  
    Gender VARCHAR (20),  
  
    FOREIGN KEY (ProductID)  
        REFERENCES Product (ProductID)  
        ON DELETE CASCADE  
);
```

-- 8. Address

CREATE TABLE Address (

AddressID INT PRIMARY KEY,

CustomerID INT NOT NULL,

HouseNo VARCHAR (30) NOT NULL,

Street VARCHAR (150) NOT NULL,

City VARCHAR (80) NOT NULL,

State VARCHAR (80) NOT NULL,

PinCode VARCHAR (10) NOT NULL,

AddressType VARCHAR (20) DEFAULT 'Home',

FOREIGN KEY (CustomerID)

REFERENCES Customer (CustomerID)

ON DELETE CASCADE

);

-- 9. Orders

CREATE TABLE Orders (

OrderID INT PRIMARY KEY

CustomerID INT NOT NULL,

AddressID INT,

OrderDate DATE NOT NULL,

OrderStatus VARCHAR (30) NOT NULL DEFAULT 'Pending',

TotalAmount DECIMAL (12,2) NOT NULL,

FOREIGN KEY (CustomerID)

REFERENCES Customer (CustomerID)

ON DELETE CASCADE,

FOREIGN KEY (AddressID)

REFERENCES CustomerAddress (AddressID)

ON DELETE SET NULL,

CHECK (TOTALAmount >= 0)

);

--10. ORDER ITEM

-- Weak Entity

-- Composite Primary Key

CREATE TABLE OrderItem (

OrderID INT NOT NULL,

ProductID INT NOT NULL,

Quantity INT NOT NULL,

UnitPrice DECIMAL(10,2) NOT NULL,

PRIMARY KEY (OrderID, ProductID),

FOREIGN KEY (OrderID)

REFERENCES Orders (OrderID)

ON DELETE CASCADE,

CHECK (Quantity > 0),

CHECK (UnitPrice >= 0)

);

-11. Payment

```
CREATE TABLE Payment (  
    PaymentID INT PRIMARY KEY,  
    OrderID INT NOT NULL UNIQUE,  
    PaymentDate DATE,  
    Amount DECIMAL (12,2) NOT NULL,  
    PaymentMethod VARCHAR(30) NOT NULL,  
    PaymentStatus VARCHAR(30) NOT NULL,  
    TransactionID VARCHAR(100) UNIQUE,  
  
    FOREIGN KEY (OrderID)  
        REFERENCES Orders (OrderID)  
        ON DELETE CASCADE  
  
    CHECK (Amount >= 0)  
);
```

-- 12. DELIVERY

```
CREATE TABLE Delivery (  
    DeliveryID INT PRIMARY KEY,  
    OrderID INT NOT NULL UNIQUE,  
    DeliveryAddressID INT,  
    DeliveryDate DATE,  
    DeliveryStatus VARCHAR(30) NOT NULL,  
    TrackingNumber VARCHAR(100) UNIQUE,
```

FOREIGN KEY (OrderID)

REFERENCES Orders (OrderID)

ON DELETE CASCADE,

FOREIGN KEY (DeliveryAddressID)

REFERENCES Address (AddressID)

ON DELETE SET NULL

);

-- 13. INSERT CUSTOMER DATA

INSERT INTO CUSTOMER

(CustomerID, FirstName, LastName, Email, RegistrationDate)

VALUES

(1, 'RAHUL', 'SHARMA', 'rahul@gmail.com', '2026-01-10'),

(2, 'Priya', 'Verma', 'priya@gmail.com', '2026-01-15'),

(3, 'Aman', 'Singh', 'aman@gmail.com', '2026-02-01'),

(4, 'Sneha', 'Gupta', 'sneha@gmail.com', '2026-02-10'),

(5, 'Arjun', 'Mehta', 'arjun@gmail.com', '2026-02-15'),

(6, 'Neha', 'Kumar', 'neha@gmail.com', '2026-03-01'),

(7, 'Rohan', 'Joshi', 'rohan@gmail.com', '2026-03-05'),

(8, 'Anjali', 'Mishra', 'anjali@gmail.com', '2026-03-10'),

(9, 'Vikas', 'Yadav', 'vikas@gmail.com', '2026-03-15'),

(10, 'Pooja', 'Agarwal', 'pooja@gmail.com', '2026-03-20');

14. INSERT CUSTOMER PHONE DATA

```
INSERT INTO CustomerPhone  
(CustomerID, PhoneNumber)
```

VALUES

(1, '9876543210'),

(1, '9123456780'),

(2, '9988776655'),

(3, '9876501234'),

(4, '9812345678'),

(5, '9898989898'),

(6, '9765432100'),

(7, '96543210000'),

(8, '954321000000'),

(9, '943210000000'),

(10, '932100000000');

15. INSERT SELLER DATA

```
INSERT INTO Seller
```

```
(SellerID, SellerName, Email, Phone, GSTIN, Rating)
```

VALUES

(101, 'TechWorld India', 'techworld@gmail.com',

'9877000001', 'GSTIN101XYZ', 4.5),

(102, 'FashionHub India', 'fashionhub@gmail.com',

'9877000003', 'GSTIN103XYZ', 4.2),

(103,

--16 Insert Category data

INSERT INTO SELLER

(SellerID, SellerName, Email, Phone, GSTIN, Rating)

VALUES

(101, 'TechWorld', 'tech@gmail.com', '9876543210', 'GST101', 4.5),

(102, 'FashionHub', 'Fashion@gmail.com', '9876543211', 'GST102', 4.2),

(103, 'HomeNeeds', 'home@gmail.com', '9876543212', 'GST103', 4.7);

--17 Insert Product data

INSERT INTO PRODUCT

(ProductID, ProductName, Price, Description, Stock, SellerID, CategoryID)

VALUES

(301, 'HP LAPTOP', 65000, 'HP Laptop', 20, 101, 201),

(302, 'HP Laptop Samsung Phone', 25000, 'Smartphone', 30, 101, 201),

(303, 'T-shirt', 799, 'Cotton T-shirt', 50, 102, 202),

(304, 'Mixer', 3499, 'Mixer Grinder', 15, 103, 203);

--18 Electronic Product

INSERT INTO ~~Electronic Product~~ ElectronicProduct

(ProductID, Brand, Model, WarrantyMonths)

VALUES

(301, 'HP', '15s', 24),

(302, 'Samsung', 'A55', 12),

(304, 'Philips', 'HL7756', 24);

--19 Clothing Product

INSERT INTO ClothingProduct

(ProductID, Size, Material, Gender)

VALUES

(303, 'L', 'Cotton', 'Unisex');

-- 20. Insert Address

INSERT INTO Address

(AddressID, CustomerID, HouseNo, Street, City, State, Pincode, AddressType)

VALUES

(401, 1, '12A', 'MG Road', 'Delhi', 'Delhi', '110001', 'Home'),

(402, 2, '22', 'Park Street', 'Kolkata', 'West Bengal', '700016', 'Home'),

(403, 3, '18', 'FC Road', 'Maharashtra', '411001', 'Home');

-- 21. Insert Orders

INSERT INTO ORDERS

(OrderID, CustomerID, AddressID, OrderDate, OrderStatus, TotalAmount)

VALUES

(501, 1, 401, '2026-08-01', 'Confirmed', 65000),

(502, 2, 402, '2026-08-02', 'Shipped', 25000),

(503, 3, 403, '2026-08-03', 'Pending', 4298);

-- 22. Insert Order Item

INSERT INTO OrderItem

(OrderID, ProductID, Quantity, UnitPrice)

VALUES

(501, 301, 1, 65000),

(502, 302, 1, 25000),

(503, 303, 1, 799),

(503, 304, 1, 3499);

-- 23. Insert Payment

Insert INTO Payment

(PaymentID, OrderID, DeliveryAddressID, DeliveryDate, DeliveryStatus, TrackingName)

VALUES

(701, 501, 401, '2026-08-05', 'Delivered', 'TRK001'),

(702, 502, 402, '2026-08-06', 'Shipped', 'TRK002');

--25. DISPLAY DATA

SELECT * FROM Saller;

SELECT * FROM Product;

SELECT * FROM Orders;

SELECT * FROM OrderItem;

SELECT * FROM Payment;

SELECT * FROM Delivery;

--26. REFERENTIAL INTEGRITY VIOLATION

INSERT INTO Orders

(OrderID, CustomerID, AddressID, OrderDate, OrderStatus, TotalAmount)

VALUES

(504, 999, 401, '2026-08-10', 'Pending', 5000);

--ERROR because CustomerID 999 does not exist

--27. ANOTHER REFERENTIAL INTEGRITY VIOLATION

INSERT INTO OrderItem

(OrderID, ProductID, Quantity, UnitPrice)

VALUES

(501, 999, 1, 5000);

--ERROR because ProductID 999 does not exist